Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination – 2020

EE1107 - ELECTRICAL PRINCIPLES – A

Answer Five question only Time : 2 hours

Question 1

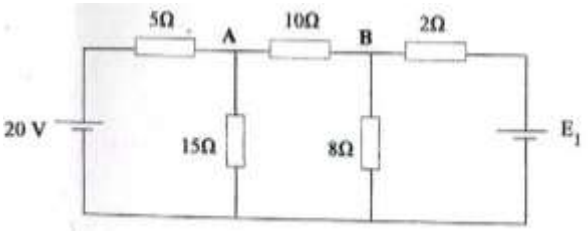

Figure Q1

The voltage between A and B is $7\mathrm{V}$ when AB is an open circuit. The resistance between A and B is 5.35Ω when AB is an open circuit and two e.m.f. sources are replaced by their internal resistances (here internal resistances are zero) in the circuit given in figure Q1.

i. Find the value of E_1 (Hint: This is an application of Thevenin's theorem). (10 marks) ii. Find the current through the 10Ω resistor in the circuit given in figure Q1 by applying the **superposition theorem**. (10 marks)

Question 2

i. Name **three (03) types of fixed capacitors**. (3 marks) ii. Draw the **charging current graph of a capacitor** and roughly mark the **time constant** on the graph. (3 marks) iii. A $1\mu\mathrm{F}$ capacitor is connected to a $50\ \mathrm{V}$ supply by closing a switch through a $2\ \mathrm{M}\Omega$ series resistor. Calculate the followings: a. The **time constant** b. **Initial charging current** c. **Voltage across the capacitor** 5 seconds after closing the switch d. The **time taken by the capacitor for charging it fully** (6 marks) iv. A parallel plate **Air capacitor is charged up to Q coulombs**. The Air gap of the same capacitor is filled with a dielectric of dielectric strength of 1.8. The energy stored in the latter capacitor is 3 times that of the aforesaid Air capacitor. Find the **charge on the dielectric-filled capacitor**. (6 marks) v. A charge of $5\mu\mathrm{C}$ is placed in an electric field experiencing a force of $2.5\mathrm{N}$. What is the **magnitude of electric field intensity**? (2 marks)

Question 3

i. Draw the direction of self-induced emf. at the inductor given in figure 02, when the value of the variable resistor is increased after closing the switch20. (2 marks)

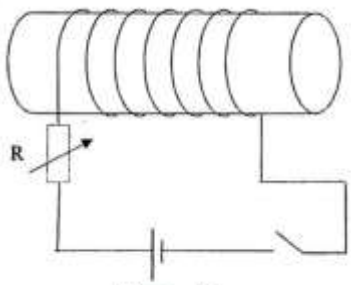

Figure - 02

ii. State the direction that a conductor is needed to be moved in a magnetic field, in order to produce the maximum emf (2 marks)

iii. State the two ways of spending the energy supply to the inductive circuit. (4 marks)

iv. Two coils having 50 and 100 turns respectively are wounded side by side on an iron core and formed a closed circuit. Cross-sectional area and mean length of the core are $100cm^2$ and 200cm respectively.

a. Estimate the mutual inductance between the coils if the relative permeability of the iron is 2000. (3 marks)

b. Find the emf induced in the second coil if a current of zero Ampere grows to 20A in a time of 0.02seconds in the 1st coil. (3 marks)

v. A steel ring with a 20cm mean radius and circular section of 2cm diameter has an Air gap of 1mm wide. It is wounded uniformly with 500 turns of wire carrying the current of 2A. If the air gap takes 70% of the total mmf, what is the reluctance of the complete magnetic circuit? (You may neglect the magnetic leakage.) (6 marks)

Question 4

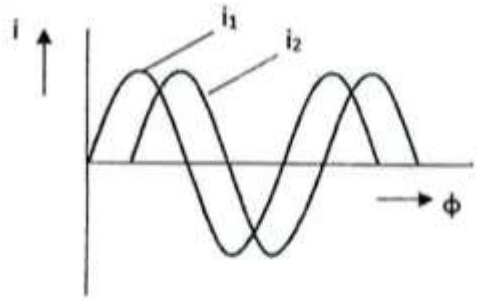

Figure – 04(i)(a)

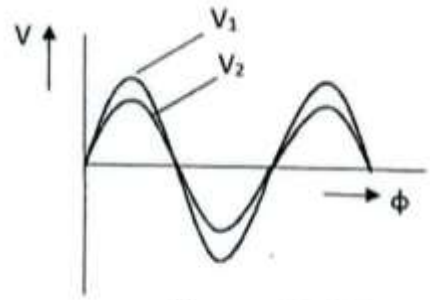

Figure – 04((i)(b)

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i. Copy the sets of waveforms given in figure Q4 (i)(a) and figure Q4 (i)(b) and mark **peak amplitudes, period, and phase difference**. Sketch the **phasor diagram** of each set of waveforms. (4 marks) ii. The frequency of the signal given below is $60\ \mathrm{Hz}$ and the peak amplitude is $5\ \mathrm{V}$. Find the **r.m.s. voltage** of the alternating voltage. Also, find the **time period for completing one cycle**. (3 marks) iii. It is found that the voltage is **leading the current by an angle** Φ° when an alternating voltage is applied to the particular circuit. Draw the **voltage triangle** and **impedance triangle**, and derive an expression for the **power factor** of the circuit. (Hint: the relationship between R and Z) (3 marks) iv. A capacitor C is connected in series with a $\mathbf{40\Omega}$ resistor across a supply of frequency $\mathbf{60}\ \mathrm{Hz}$. A current of $\mathbf{3}\mathrm{A}$ flows and the circuit impedance is $\mathbf{50\Omega}$. Calculate the following: a. The **value of the capacitor** b. The **supply voltage** c. The **phase angle** between the supply voltage and current d. The **potential difference across the resistor** e. The **potential difference across the capacitor** f. Draw the **phasor diagram** (6 marks) v. A coil having a resistance of 10Ω and an inductance of $125\ \mathrm{mH}$ is connected in series with a $60\mu\mathrm{F}$ capacitor across a $120\ \mathrm{V}$ supply. At what frequency does the **resonance occur**? Find the **current through the circuit at this frequency**. (4 marks)

Question 5

Two impedances given by $Z_1 = (4 + j3)\Omega$ and $Z_2 = (8 - j6)\Omega$ are joined in **parallel** and connected across a voltage of magnitude $\mathbf{200}\mathrm{V}$ and angle $\mathbf{0}^\circ$ (i.e. $200\angle 0^\circ$). Calculate:

i. **Circuit current** (6 marks) ii. **Current in each branch** (6 marks) iii. **Average power** delivered to the circuit (2 marks) iv. **Power taken by each branch** (2 marks) v. **Power factor** of individual circuits and combination (4 marks)

Question 6

i. Write down **two advantages of a three-phase power system** over a single-phase power system. (2 marks) ii. What is meant by **'balanced load'** in a three-phase power system? (2 marks) iii. State the possible **phase sequences** in a three-phase electrical power system if the three phases are R, Y, B? (3 marks) iv. Mark the **line voltage and phase voltage in a three-phase star connection** with the help of a diagram. (3 marks) v. An impedance of $(12 + j16)\Omega$ balance three-phase **delta connected load** consumes $4.8\ \mathrm{kW}$ per phase. The line current drawn by the load is $34.64\mathrm{A}$. Find the **line voltages**. (5 marks) vi. A **star-connected load** consists of three (03) identical coils each of resistance 30Ω and inductance $127.3\ \mathrm{mH}$. When the line current is $5.08\mathrm{A}$ and the supply frequency is $50\ \mathrm{Hz}$, calculate the **line voltage**. (5 marks)